

THE AESTHETICS OF RANDOMNESS

Mandelbrot's library—Was Galileo blind?—Pearls to swine—Self-affinity—How the world can be complicated in a simple way, or, perhaps, simple in a very complicated way

THE POET OF RANDOMNESS

It was a melancholic afternoon when I smelled the old books in Benoît Mandelbrot's library. This was on a hot day in August 2005, and the heat exacerbated the musty odor of the glue of old French books bringing on powerful olfactory nostalgia. I usually succeed in repressing such nostalgic excursions, but not when they sneak up on me as music or smell. The odor of Mandelbrot's books was that of French literature, of my parents' library, of the hours spent in bookstores and libraries when I was a teenager when many books around me were (alas) in French, when I thought that Literature was above anything and everything. (I haven't been in contact with many French books since my teenage days.) However abstract I wanted it to be, Literature had a physical embodiment, it had a smell, and this was it.

The afternoon was also gloomy because Mandelbrot was moving away, exactly when I had become entitled to call him at crazy hours just because I had a question, such as why people didn't realize that the 80/20 could be 50/01. Mandelbrot had decided to move to the Boston area, not to retire, but to work for a research center sponsored by a national laboratory. Since he was moving to an apartment in Cambridge, and leaving his oversize house in the Westchester suburbs of New York, he had invited me to come take my pick of his books.

Even the titles of the books had a nostalgic ring. I filled up a box with French titles, such as a 1949 copy of Henri Bergson's *Matière et mémoire*, which it seemed Mandelbrot bought when he was a student (the smell!).

After having mentioned his name left and right throughout this book, I will finally introduce Mandelbrot, principally as the first person with an academic title with whom I ever spoke about randomness without feeling defrauded. Other mathematicians of probability would throw at me theorems with Russian names such as "Sobolev," "Kolmogorov," Wiener measure, without which they were lost; they had a hard time getting to the heart of the subject or exiting their little box long enough to consider its empirical flaws. With